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Stereoselective Iodolactonization of 4-Allenoic Acids with Efficient Chirality Transfer: Development of a New Electrophilic Iodination Reagent

Xiaobing Zhang,[a] Chunling Fu,[a] Yihua Yu,[b] and Shengming Ma*[a]

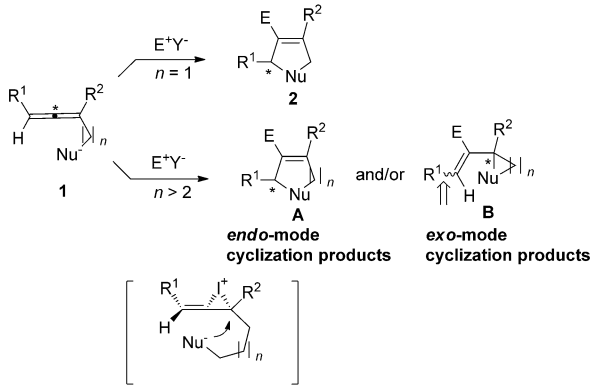
Abstract: A highly stereoselective iodolactonization of 4-allenoic acids with a new sterically demanding electrophilic iodination reagent to afford optically active γ-butyrolactones has been developed. The reaction shows high efficiency of axial chirality transfer and excellent *Z*/*E* selectivity and has been applied to the synthesis of chiral *cis*-β,γ-disubstituted γ-butyrolactones to give very high diastereomeric and enantiomeric excess values. The reaction has been successfully utilized in the synthesis of naturally occurring compounds (+)-*cis*-whisky lactone and (+)-*cis*-3-methyl-4-decanolide.

Keywords: allenes · chirality · cyclization · electrophilic addition · iodine

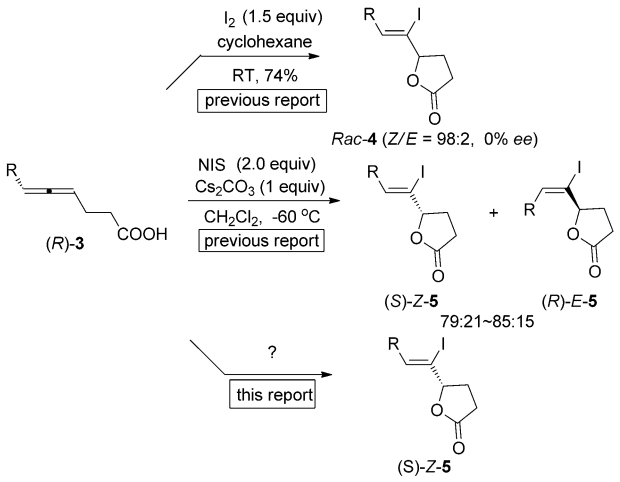
Introduction

Electrophilic cyclization of 2-allenyl compounds with a nucleophilic moiety at the 1-position has proved to be a powerful method for the synthesis of five-membered cyclic compounds.[1,2] The intrinsic axial chirality of the allene moiety permits enantioselective synthesis of optically active cyclic products provided that chirality transfer may be realized through an *anti*-cyclization mode.[3] However, with an elongated carbon chain between the allene unit and the nucleophilic moiety (compounds **1**; $n>2$), in addition to the issues of regioselectivity and axial chirality transfer, it is a challenge to control to the configuration of the remaining C=C bond in products **B**, which are obtained through an *exo*-cyclization mode (Scheme 1).[4]

In 2007, we observed that the iodolactonization of (*R*)-4-allenoic acids **3** affords racemic (*Z*)-**5** by way of a cationic mechanism, whereas electrophilic *anti* cyclization afforded an inseparable mixture of (*S*)-(*Z*)-**5** and (*R*)-(*E*)-**5** (Scheme 2). It should be noted that the absolute configurations in these two products are opposite. To make this type of cyclization useful in organic synthesis, we must address



Scheme 1. Electrophilic *anti* cyclization with chirality transfer (Nu = nucleophile).



Scheme 2. Two different mechanisms for iodolactonization of (*R*)-4-allenoic acids (NIS = *N*-iodosuccinimide).

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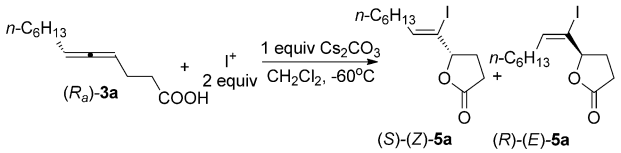
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such an issue.[4b] Herein, we report the realization of an iodolactonization with excellent *Z* selectivity and efficient chirality transfer with the help of a new electrophilic iodination reagent **8**. The protocol has been applied to the synthesis of β,γ-disubstituted *cis* lactones and two naturally occurring *cis* lactones, (+)-*cis*-whisky lactone and (+)-*cis*-3-methyl-4-decanolide.

Results and Discussion

Cyclization to address *Z/E* selectivity and maintain high efficiency for axial chirality transfer: To develop our new protocol, we used (R_a)-**3a** with allene axial chirality as the model substrate. As expected, albeit with much disappointment, (*S*)-5-(1′-iodo-1′-(*Z*)-octenyl)-4,5-dihydro-2(3*H*)-furanone [(*S*)-(*Z*)-**5a**] and (*R*)-5-(1′-iodo-1′-(*E*)-octenyl)-4,5-dihydro-2(3*H*)-furanone [(*R*)-(*E*)-**5a**] were obtained with a diastereomeric ratio (d.r.) of 78:22, in 99% yield when NIS (**6**; 2 equiv) was used, although there is no obvious racemization (Table 1, entry 2).

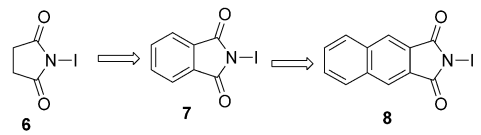
Table 1. Effect of the I^+ source on the iodolactonization of γ-allenoic acid (*R*)-**3a**.



Entry	I^+	*t* [h]	Yield of **5a** [%]	(*S*)-(*Z*)-**5a**/(*R*)-(*E*)-**5a**	*ee* [%] (*S*)-(*Z*)-**5a**	*ee* [%] (*R*)-(*E*)-**5a**
1[a]	**6**	13	86	78:22	98	98
2	**6**	18	99	78:22	98	97
3	**7**	16	71	93:7	98	96
4	**8**	12	85	97:3	98	97

[a] Reaction performed without Cs_2CO_3.

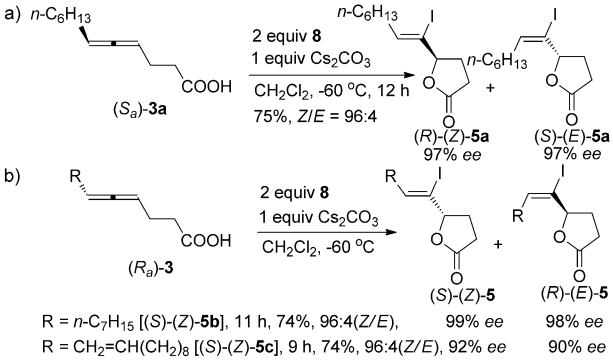
It should be noted that the configuration of the remaining C=C bond and the absolute configuration of the central chirality in these two isomers are opposite to each other,[4b] which also results in an enantiomeric excess (*ee*) of the product of only 56% after hydrogenation of the C=C bond and reduction of the C–I bond. By consideration of the nature of this iodolactonization reaction, we reasoned that a more sterically demanding electrophilic iodination reagent, such as **7** or **8**, might provide a much-improved stereoselectivity (Scheme 3).



Scheme 3. Concept for new iodination reagents with improved stereoselectivity that maintain the *anti* nature of the cyclization.

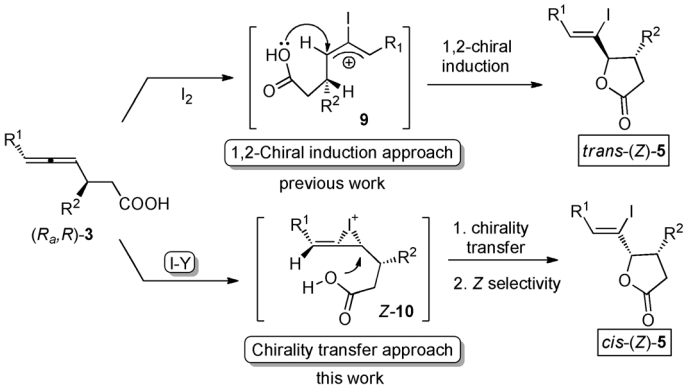
First, when **7**—with a benzene ring to replace the two methylene units of **6**—was prepared and used,[5] the ratio of (*S*)-(*Z*)-**5a**/(*R*)-(*E*)-**5a** was improved to 93:7 (Table 1, entry 3). To our delight, when **8**—with a naphthalene ring modification[6]—was used, the selectivity of (*S*)-(*Z*)-**5a**/(*R*)-(*E*)-**5a** was improved further to 97:3 (Table 1, entry 4).

Such a strategy is general for the highly stereoselective iodolactonization of 4-allenoic acids with axial chirality. The efficiency of chirality transfer is very high, demonstrated by the reactions of (R_a)-**3b**, and (R_a)-**3c** (Scheme 4b). As expected, enantiomer (*R*)-(*Z*)-**5a** may be prepared from (S_a)-**3a** (Scheme 4a).



Scheme 4. Synthesis of optically active γ-substituted butyrolactones by iodolactonization of 4-allenoic acids with chirality transfer.

Application to the synthesis of thermodynamically unstable β,γ-disubstituted *cis* butyrolactones: We have previously reported[4b,f] that the highly diastereoselective iodolactonization of optically active 4-allenoic acids (R_a,*R*)-**3**, via the intermediacy of π-allylic cationic species **9**, led to the formation of thermodynamically more stable β,γ-disubstituted *trans* products by 1,2-chiral induction (Scheme 5). Therefore, we envisioned that it may now be possible to synthesize the relatively unstable β,γ-disubstituted *cis*-γ-butyrolactones with high stereoselectivity, provided that the newly formed chiral carbon center at the γ-position is generated by chirali-



Scheme 5. Concept for the synthesis of *cis* lactones.

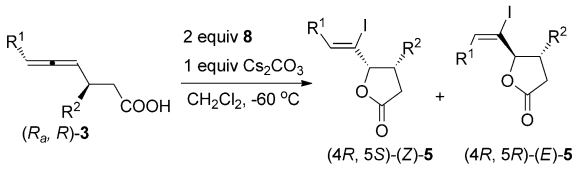
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ty transfer from the allene moiety with an *R* configuration, instead of by 1,2-chiral induction.

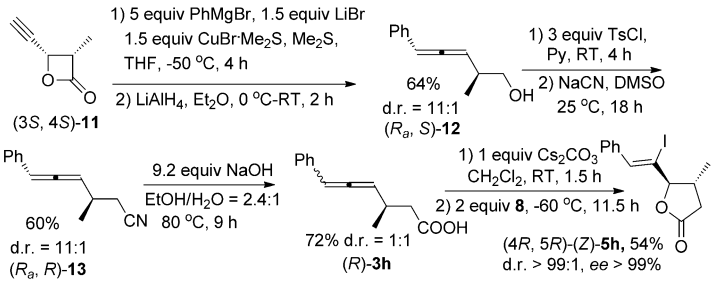
We were happy to see that chirality transfer is operational for the synthesis of thermodynamically disfavored β,γ-disubstituted *cis* butyrolactones; the reaction of (R_a,*R*)-(−)-**3d** with **8** afforded (4*R*,5*S*)-5-(1′-iodo-1′-(*Z*)-butenyl)-4-methyl-4,5-dihydro-2(3H)-furanone [(4*R*,5*S*)-(*Z*)-**5d**] in 70% yield with a *E*/*Z* ratio of >97:3 (Table 2, entry 1). An allyl group was also tolerated at the β position of the furanone, which indicated a much higher reactivity of the allene moiety relative to the allylic C=C bond (Table 2, entries 3 and 4). Furthermore, an R^1 substituent with larger steric hindrance, such as an isopropyl group, could also be used in the reaction and gave excellent selectivity (Table 2, entry 5).

Table 2. Iodolactonization of 4-allenoic acids **3** for the synthesis of *cis*-B,γ-disubstituted lactones **5**.



Entry	**3**	R^1	R^2	*t* [h]	*Z*-**5**	Yield of (*Z*)-**5** [%]	(*Z*)-**5**/(*E*)-**5**	*ee* of (*Z*)-**5** [%]
1	(R_a,*R*)-**3d**	C_2H_5	CH_3	10	(4*R*,5*S*)-**5d**	70	>97:3	99
2	(R_a,*R*)-**3e**	*n*-C_4H_9	CH_3	8	(4*R*,5*S*)-**5e**	71	>98:2	99
3	(R_a,*R*)-**3f**	*n*-C_4H_9	allyl	14	(4*R*,5*S*)-**5f**	83	>99:1	>99
4	(S_a,*S*)-**3f**	*n*-C_4H_9	allyl	10	(4*S*,5*R*)-**5f**	88	>99:1	>99
5	(S_a,*S*)-**3g**	*i*-C_3H_7	CH_3	11	(4*S*,5*R*)-**5g**	68	>99:1	99

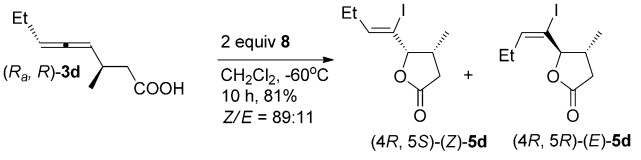
To further illustrate the generality of this strategy, (*R*)-3-methyl-6-phenylhexa-4,5-dienoic acid [(*R*)-**3h**] was prepared by the chemistry shown in Scheme 6.[4b] Compound (3*S*,4*S*)-3-methyl-4-ethynyloxetan-2-one [(3*S*,4*S*)-**11**] was converted to a 3-allenoic acid, followed by reduction with $LiAlH_4$ to form alcohol (R_a,*S*)-**12**. Tosylation and subsequent cyanation afforded nitrile (R_a,*R*)-**13**. Finally, hydrolysis with NaOH in EtOH/H_2O afforded the allenoic acid (*R*)-**3h**. However, epimerization of the axial chirality of the allene moiety was observed, which led to the formation of a pair of equal diastereoisomers. When (*R*)-**3h** was treated with **8** under the standard conditions, (4*R*,5*R*)-(*Z*)-**5h** was afforded as a single isomer in 54% yield. This result indicated that the reaction proceeded through the intermediacy of π-allylic cationic species **9**, which led to the formation of the thermodynamically more stable *trans* product by 1,2-chiral induction.[4b] This is clearly caused by stabilization of the cationic intermediate by the phenyl substituent.



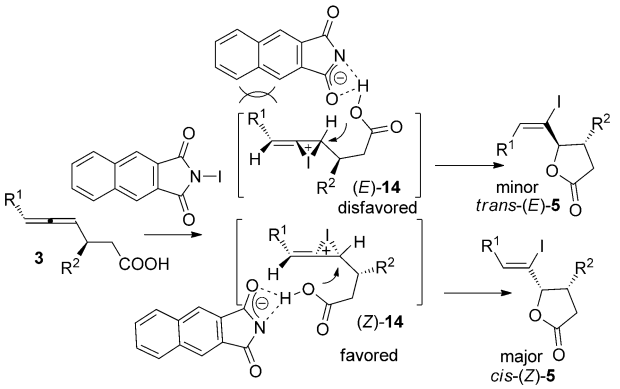
Scheme 6. Synthesis and iodolactonization of (*R*)-**3h** (TsCl = tosyl chloride, Py = pyridine).

Rationale for the observed stereoselectivity: It is interesting to observe that in the absence of Cs_2CO_3, the reaction of (R_a,*R*)-**3d** with **8** afforded a much lower ratio (89:11) of (4*R*,5*S*)-(*Z*)-**5d**/(4*R*,5*R*)-(*Z*)-**5d** (Scheme 7), which indicates the unique effect of the base on the stereoselectivity of this reaction.

Based on these facts, the high *Z* stereoselectivity and efficiency of chirality transfer observed can be rationalized as follows (Scheme 8). The reaction of allenoic acid **3** with iodination reagent **8** would afford iodonium intermediate (*Z*)-**14** or (*E*)-**14**, in which the amide anion plays an important role (formation of a hydrogen-bonded four-membered unit)[7] in determining the stereoselectivity by the steric interaction with R^1.



Scheme 7.



Scheme 8. Rationale for the high *Z*/*E* selectivity observed in the iodocyclization with highly efficient chirality transfer.

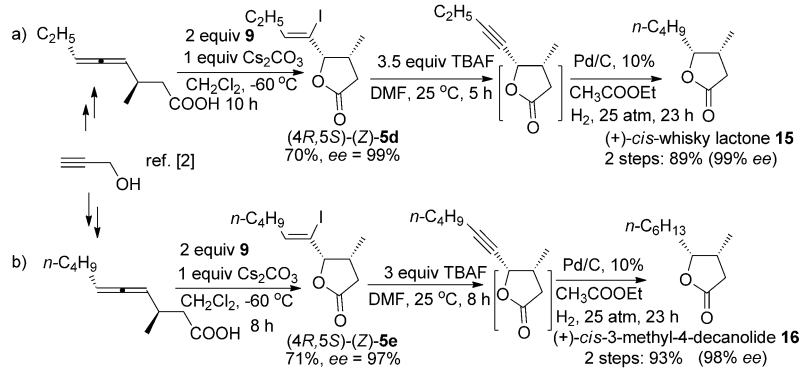
Application to the synthesis of naturally occurring lactones: (+)-*Cis*-whisky lactone (**15**) was extracted from wood into alcoholic beverages during fermentation and/or maturation;[8] (+)-*cis*-3-methyl-4-decanolide (**16**) exists in the vola-

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tile components of the juice of blood and blond oranges, and has been used as an active ingredient in oral bactericides in dentifrices.[9] Thus, there are many reports devoted to the synthesis of **15**[10] and **16**.[11] Fukuzawa et al.,[10b] Procter et al.,[10c] and Dai et al.[10d] reported SmI_2-catalyzed cyclizations of α,β-unsaturated esters with ketones or aldehydes to prepare **15** in 94 (d.r. = 97:3), 91, and 99% *ee* (d.r = 90:10), respectively. Ghosh et al.[10e] and Elsey et al.[10f] reported the synthesis of **15**, in which the chiral centers of the target were constructed by the reaction of a chiral reagent with a key intermediate to afford a pair of diastereoisomers, which were separated by chromatography on silica gel; subsequent transformations gave the target with 6.6 (nine steps) and 6.2% yield (six steps), respectively. For the synthesis of **16**, Wu et al.[11] reported a route from (*S*)-*N*-propionyl-4-benzyl-3-propanoyloxazolidin-2-one and heptanal in five steps, with no *ee* value given. Kitahara et al. reported a synthesis of **16** in eight steps from (1*S*,5*R*)-2-oxabicyclo[3.3.0]-oct-6-en-3-one, however, the yield was low (2%), due to the fact that two steps in this approach gave a larger number of recovery.[9] Syntheses of **15** and **16** from chiral auxiliaries gave good yields but low d.r. and *ee* values,[10b–d] whereas syntheses that gave good stereoselectivity of the target molecules were accompanied by low yields[10e–f] and advanced chiral starting materials[9,11] were required. Therefore, development of new efficient routes with higher stereoselectivities and yields, from simple starting materials, is still of interest.

The synthetic application of our iodolactonization reaction to short asymmetric syntheses of **15** and **16** was demonstrated (Scheme 9). Both **15** and **16** were synthesized easily in two steps, from (4*R*,5*S*)-(*Z*)-**5d** (89%) and (4*R*,5*S*)-(*Z*)-**5e** (93%), respectively. Measured optical rotation and ^1H and ^{13}C NMR analysis of the products were in good agreement with previous reports. Compared to all of the known routes, this process displays the highest diastereoselectivity and *ee* for the final products.[10,11]



Scheme 9. Application to the synthesis of naturally occurring *cis*-γ-lactones (TBAF = tetrabutylammonium fluoride).

Conclusion

We have developed a highly stereoselective iodolactonization of 4-allenoic acids, which proceeds via a three-membered iodonium ion intermediate to afford chiral γ-butyrolactones, by development of a sterically demanding iodination reagent. The synthesis of β,γ-disubstituted *cis*-γ-butyrolactones from the starting materials with different steric nature is especially noteworthy. This reaction has been successfully applied to the synthesis of naturally occurring lactones, (+)-*cis*-whisky lactone and (+)-*cis*-3-methyl-4-decanolide. Due to the high *Z*/*E* stereoselectivity, excellent efficiency of chirality transfer, the substituent-loading capability of the starting allenoic acids, and potential uses of the products, this reaction—in particular, the new iodination reagent—will be useful in organic chemistry, especially for the synthesis of a series of naturally occurring *cis*-γ-butyrolactones.[12] Further studies in this area are ongoing in our laboratory.

Experimental Section

Materials: Cs_2CO_3 was purchased from Alfa. Other commercially available chemicals were used without additional purification unless otherwise noted. ^1H and ^{13}C NMR experiments were performed in $CDCl_3$ and referenced to the signals of tetramethylsilane ($\delta = 0$ ppm) and residual chloroform (77.0 ppm), respectively.

Preparation of *N*-iodo-2,3-naphthalenedicarboximide (8):[5] A solution of KOH (0.7399 g, 13.2 mmol) in EtOH (75% aq, 3 mL) was added to a solution of 2,3-naphthalenedicarboximide (2.3306 g, 11.8 mmol) in EtOH (40 mL) at 80 °C. The reaction mixture was stirred at 80 °C for 10 min, then filtered. The residue was washed with acetone and dried under vacuum to afford the crude product, which was used immediately in the next step.

Potassium 2,3-naphthalenedicarboximide (prepared above), bis(trifluoroacetoxy)iodobenzene (2.5843 g, 6.0 mmol), and MeCN (55 mL) were combined then stirred for 14 h. The suspension was filtered, the residue washed with MeCN, and dried under vacuum. The product was used in the next step without further purification.

Phenyliodine(III) bis(2,3-naphthalenedicarboximidate) [prepared above], CCl_4 (100 mL), and iodine (3.0462 g, 12.0 mmol) were combined then stirred at RT for 23 h. The mixture was filtered and the residue washed with Et_2O to afford **8** (3.2196 g, including 2,3-naphthalenedicarboximide (2%), 84% over three steps). The product could be stored under an air atmosphere at −20 °C for >6 months without observable decomposition. Solid; ^1H NMR (300 MHz, $(CD_3)_2SO$): $\delta = 8.41$ (s, 2H; Ar–H), 8.30–8.17 (m, 2H; Ar–H), 7.82–7.70 ppm (m, 2H; Ar–H); ^{13}C NMR (75 MHz, $(CD_3)_2SO$): $\delta = 171.4$, 135.5, 131.1, 130.1, 129.5, 124.9 ppm; IR (neat): $\tilde{\nu} =$ 1742, 1680, 1605, 1515, 1451, 1422, 1337, 1291, 1146, 1104, 1063 cm^{-1}; MS: *m*/*z* (%): 323 $[M]^+$ (3.39), 126 (100); HRMS: *m*/*z* calcd for $C_{12}H_6INO_2$: 322.9443 $[M]^+$; found: 322.9449.

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Preparation of optically active 4-allenoic acids (3)

*Method a [(R_a)-(−)-**3a**]*: *p*-TsCl (42.2949 g, 0.23 mol) was added in six portions to a solution of (R_a)-undeca-3,4-dien-1-ol (12.0510 g, 0.072 mol, synthesized from (*R*)-nonyn-3-ol,[4b] 99% *ee*) in pyridine (90 mL) at 0°C within 30 min. The reaction mixture was stirred at RT for 5 h, then quenched with water (20 mL) and extracted with Et_2O (3×80 mL). The combined organic layers were washed with brine (×3), dried over anhydrous Na_2SO_4, filtered, and evaporated under vacuum. The tosylate was used in the next step without further purification.

NaCN (5.4104 g, 0.11 mol) was added to a solution of the crude tosylate (prepared above) in DMSO (30 mL). The reaction mixture was stirred for 18 h at 20°C, quenched with H_2O (20 mL), and extracted with Et_2O (3×100 mL). The combined organic layers were washed with brine (×2), dried over Na_2SO_4, filtered, evaporated under vacuum, and purified by chromatography on silica gel (petroleum ether/EtOAc = 100:1) to afford the product, which was used directly in the next step.

A solution of NaOH (20.0013 g, 0.50 mol) in H_2O (26 mL) at 80°C was added to a solution of dodeca-4,5-dienenitrile (prepared above) in EtOH (75 mL). The reaction mixture was stirred at 80°C for 5 h, the solvent was evaporated under vacuum, and the residue was quenched with water (100 mL). The aqueous solution was washed with Et_2O, then acidified with HCl (5% aq) to pH≈1, and extracted with Et_2O (3×100 mL). The combined organic layers were washed with water and brine, dried over anhydrous Na_2SO_4, filtered, and concentrated under vacuum. Chromatography of the crude product on silica gel (petroleum ether/EtOAc = 5:1) afforded (R_a)-(−)-**3a** (10.3239 g, 73% over 3 steps). Liquid; $[\alpha]_D^{20} = -71.6$ ($c = 0.97$ in $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 11.17$ (brs, 1H; COOH), 5.20–5.11 (m, 2H; 2×CH=), 2.53–2.43 (m, 2H; CH_2), 2.36–2.24 (m, 2H; CH_2), 2.03–1.90 (m, 2H; CH_2), 1.44–1.20 (m, 8H; 4×CH_2), 0.88 ppm (t, $J = 6.8$ Hz, 3H; CH_3).

*Compound (S_a)-(+)-**3a***: Following method a, the reaction of (S_a)-undeca-3,4-dien-1-ol (1.4901 g, 8.98 mmol, synthesized from (*S*)-nonyn-3-ol, 99% *ee*) and *p*-TsCl (5.2027 g, 27.24 mmol, added in two portions) in pyridine (10 mL), afforded the tosylate, which was used in the next step without further purification.

The reaction of the tosylate (prepared above) and NaCN (0.6631 g, 13.53 mmol) in DMSO (11 mL) afforded dodeca-4,5-dienenitrile after chromatography (petroleum ether/EtOAc = 60:1), which was used directly in the next step.

The reaction of dodeca-4,5-dienenitrile (prepared above) with NaOH (2.9622 g, 74.06 mmol) in EtOH (15 mL) and water (6 mL) afforded (S_a)-(+)-**3a** (1.2882 g, 73% over three steps) after chromatography (petroleum ether/EtOAc = 20:1–5:1). Liquid; $[\alpha]_D^{20} = +69.8$ ($c = 1.12$ in $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 11.42$ (brs, 1H; COOH), 5.21–5.10 (m, 2H; 2×CH=), 2.54–2.43 (m, 2H; CH_2), 2.36–2.25 (m, 2H; CH_2), 2.03–1.92 (m, 2H; CH_2), 1.44–1.21 (m, 8H; 4×CH_2), 0.88 ppm (t, $J = 6.6$ Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 203.6$, 179.8, 92.9, 89.2, 33.1, 31.7, 29.1, 28.8, 23.5, 22.6, 14.1 ppm; IR (neat): $\tilde{\nu} = 2953$, 2926, 2856, 1964, 1712, 1435, 1278, 1250, 1211, 1173 cm^{-1}; MS: *m*/*z* (%): 196 $[M]^+$ (0.61), 126 (100); HRMS: *m*/*z* calcd for $C_{12}H_{20}O_2$: 196.1463 $[M]^+$; found: 196.1461.

*Compound (R_a)-(−)-**3b***: Following method a, the reaction of (R_a)-dodeca-3,4-dien-1-ol (0.7966 g, 4.38 mmol, synthesized from (*R*)-decayn-3-ol, >99% *ee*) and *p*-TsCl (2.5453 g, 13.33 mmol, added in two portions) in pyridine (6 mL), afforded the tosylate, which was used in the next step without further purification.

The reaction of the tosylate (prepared above) and NaCN (0.3213 g, 6.56 mmol) in DMSO (4 mL) afforded trideca-4,5-dienenitrile after chromatography (petroleum ether/EtOAc = 100:1), which was used directly in the next step.

The reaction of trideca-4,5-dienenitrile (prepared above) with NaOH (1.5234 g, 38.1 mmol) in EtOH(6 mL) and water (2 mL) afforded (R_a)-(−)-**3b** (0.5591 g, 61% over three steps) after chromatography (petroleum ether/EtOAc = 20:1). Liquid; $[\alpha]_D^{20} = -61.0$ ($c = 1.00$ in $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 9.77$ (brs, 1H; COOH), 5.20–5.11 (m, 2H; 2×CH=), 2.53–2.43 (m, 2H; CH_2), 2.36–2.24 (m, 2H; CH_2), 2.02–1.90 (m, 2H; CH_2), 1.44–1.19 (m, 10H; 5×CH_2), 0.88 ppm (t, $J = 6.6$ Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 203.6$, 179.6, 92.9, 89.2, 33.1, 31.9, 29.1, 28.8, 23.5, 22.7, 14.1 ppm; IR (neat): $\tilde{\nu} = 2926$, 2855, 1963, 1712, 1434, 1336, 1280, 1250, 1211, 1172 cm^{-1}; MS: *m*/*z* (%): 210 $[M]^+$ (0.38), 126 (100); elemental analysis calcd (%) for $C_{13}H_{22}O_2$: C 74.24, H 10.54; found: C 74.05, H 10.84.

*Compound (R_a)-(−)-**3c***: Following method a, the reaction of (R_a)-pentadeca-3,4,14-trien-1-ol (1.1863 g, 5.34 mmol, synthesized from (*R*)-tridec-12-en-1-yn-3-ol, 96% *ee*) and *p*-TsCl (3.1329 g, 16.40 mmol, added in two portions) in pyridine (6 mL) afforded the tosylate, which was used in the next step without further purification.

The reaction of the tosylate (prepared above) and NaCN (0.4501 g, 11.25 mmol) in DMSO (10 mL) afforded hexadeca-4,5,15-trienenitrile after chromatography (petroleum ether/EtOAc = 100:1–40:1), which was used directly in the next step.

The reaction of hexadeca-4,5,15-trienenitrile (prepared above) with NaOH (2.2006 g, 55.02 mmol) in EtOH (8 mL) and water (3 mL) afforded (R_a)-(−)-**3c** (0.9852 g, 74% over three steps) after chromatography (petroleum ether/EtOAc = 20:1–3:1). Liquid; $[\alpha]_D^{20} = -52.5$ ($c = 1.01$ in $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 9.95$ (brs, 1H; COOH), 5.88–5.73 (m, 1H; CH=), 5.20–5.10 (m, 2H; 2×CH=), 5.03–4.89 (m, 2H; CH_2=), 2.48 (t, $J = 7.2$ Hz, 2H; CH_2), 2.35–2.24 (m, 2H; CH_2), 2.09–1.91 (m, 4H; 2×CH_2), 1.44–1.23 ppm (m, 12H; 6×CH_2); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 203.6$, 179.6, 139.2, 114.1, 92.9, 89.2, 33.8, 33.1, 29.5, 29.4, 29.1, 28.9, 28.8, 23.5 ppm; IR (neat): $\tilde{\nu} = 2926$, 2854, 1963, 1712, 1640, 1437, 1336, 1279, 1211, 1173 cm^{-1}; MS (ESI): *m*/*z*: 250 $[M]^+$, 249 $[M-H]^+$, 295 $[M+HCOOH]^+$; HRMS: *m*/*z* calcd for $C_{16}H_{25}O_2$: 249.1860 $[M-H]^+$; found: 249.1859.

*Compound (R_a,*R*)-(−)-**3d***: Following method a, the reaction of (R_a,*S*)-2-methylhepta-3,4-dien-1-ol (0.9579 g, 7.60 mmol), and *p*-TsCl (4.4276 g, 23.18 mmol, added in three portions) in pyridine (10 mL) afforded the tosylate, which was used in the next step without further purification.

The reaction of the tosylate (prepared above) and NaCN (0.5601 g, 11.43 mmol) in DMSO (8 mL) afforded 3-methylocta-4,5-dienenitrile after chromatography (petroleum ether/Et_2O = 100:1), which was used directly in the next step.

The reaction of 3-methylocta-4,5-dienenitrile (prepared above) with NaOH (2.6315 g, 65.79 mmol) in EtOH (10 mL) and water (3 mL) afforded (R_a,*R*)-(−)-**3d**[4f] (0.5699 g, 49% over three steps) after chromatography (petroleum ether/EtOAc = 10:1). Liquid; $[\alpha]_D^{20} = -74.0$ ($c = 1.00$ in $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 11.47$ (brs, 1H; COOH), 5.30–5.14 (m, 2H; 2×CH=), 2.75–2.59 (m, 1H; CH), 2.51–2.39 (m, 1H; one proton of CH_2), 2.36–2.24 (m, 1H; one proton of CH_2), 2.06–1.93 (m, 2H; CH_2), 1.09 (d, $J = 6.9$ Hz, 3H; CH_3), 0.99 ppm (t, $J = 7.4$ Hz, 3H; CH_3).

*Compound (R_a,*R*)-(−)-**3e***: Following method a, the reaction of (R_a,*S*)-2-methylnona-3,4-dien-1-ol (0.9543 g, 6.20 mmol) and *p*-TsCl (3.5935 g, 18.81 mmol, added in two portions) in pyridine (6 mL) afforded the tosylate, which was used in the next step without further purification.

The reaction of the tosylate (prepared above) and NaCN (0.4567 g, 9.32 mmol) in DMSO (6 mL) afforded 3-methyldeca-4,5-dienenitrile after chromatography(petroleum ether/EtOAc = 80:1), which was used directly in the next step.

The reaction of 3-methylocta-4,5-dienenitrile (prepared above) with NaOH (2.1011 g, 52.53 mmol) in EtOH (5 mL) and water (2 mL) afforded (R_a,*R*)-(−)-**3e** (0.7890 g, 70% over three steps) after chromatography (petroleum ether/EtOAc/CH_2Cl_2 = 20:1:1–10:1:1). Liquid; $[\alpha]_D^{20} = -66.4$ ($c = 1.05$ in $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 10.27$ (brs, 1H; COOH), 5.24–5.10 (m, 2H; 2×CH=), 2.74–2.58 (m, 1H; CH), 2.52–2.38 (m, 1H; one proton of CH_2), 2.36–2.24 (m, 1H; one proton of CH_2), 2.26–1.90 (m, 2H; CH_2), 1.44–1.25 (m, 4H; 2×CH_2), 1.08 (d, $J = 6.9$ Hz, 3H; CH_3), 0.89 ppm (t, $J = 6.9$ Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 202.4$, 179.4, 95.5, 93.4, 41.2, 31.3, 29.8, 28.6, 22.2, 20.3, 13.9 ppm; IR (neat): $\tilde{\nu} = 2961$, 2929, 2873, 1962, 1710, 1458, 1410, 1294, 1231, 1199 cm^{-1}; MS: *m*/*z* (%): 182 $[M]^+$ (1.94), 140 (100), HRMS: *m*/*z* calcd for $C_{11}H_{18}O_2$: 182.1307 $[M]^+$; found: 182.1308.

*Compound (R_a,*R*)-(−)-**3f***: Following method a, the reaction of (R_a,*S*)-2-allylnona-3,4-dien-1-ol (1.4171 g, 7.87 mmol) and *p*-TsCl (4.4995 g,

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23.56 mmol, added in two portions) in pyridine (8 mL) afforded the tosylate, which was used in the next step without further purification.

The reaction of the tosylate (prepared above) and NaCN (1.1821 g, 24.12 mmol) in DMSO (8 mL) afforded 3-allyldeca-4,5-dienenitrile after chromatography (petroleum ether/EtOAc = 40:1), which was used directly in the next step.

The reaction of 3-allylocta-4,5-dienenitrile (prepared above) with NaOH (3.1441 g, 78.60 mmol) in EtOH (15 mL), and water (6 mL) afforded (R_a,R)-(−)-**3f** (1.2208 g, 75% over three steps) after chromatography (petroleum ether/EtOAc = 20:1–5:1). Liquid; $[\alpha]_D^{20} = -65.5$ ($c = 1.09$ in $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 10.99$ (brs, 1H; COOH), 5.87–5.68 (m, 1H; CH=), 5.25–4.95 (m, 4H; =CH_2 and 2×CH=), 2.72–2.56 (m, 1H; CH), 2.50–2.30 (m, 2H; CH_2), 2.30–2.10 (m, 2H; CH_2), 2.04–1.88 (m, 2H; CH_2), 1.45–1.25 (m, 4H; 2×CH_2), 0.90 ppm (t, $J = 6.5$ Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 202.9$, 179.4, 135.7, 117.1, 93.5, 93.4, 39.1, 38.5, 34.5, 31.2, 28.6, 22.2, 13.9 ppm; IR (neat): $\tilde{\nu} = 2959$, 2928, 1963, 1709, 1641, 1437, 1379, 1297, 1238, 1181 cm^{-1}; MS (ESI): m/z: 208 $[M]^+$, 207 $[M-H]^+$; HRMS: m/z calcd for $C_{13}H_{19}O_2$: 207.1391 $[M-H]^+$; found: 207.1398.

Compound (S_a,S)-(+)-***3f***: Following method a, the reaction of (S_a,R)-2-allylnona-3,4-dien-1-ol (1.1045 g, 6.14 mmol) and *p*-TsCl (3.4796 g, 18.22 mmol, added in two portions) in pyridine (6 mL) afforded the tosylate, which was used in the next step without further purification.

The reaction of the tosylate (prepared above) and NaCN (0.5911 g, 12.06 mmol) in DMSO (6 mL) afforded 3-allyldeca-4,5-dienenitrile after chromatography (petroleum ether/Et_2O = 80:1–40:1), which was used directly in the next step.

The reaction of 3-allylocta-4,5-dienenitrile (prepared above) with NaOH (2.1215 g, 53.04 mmol) in EtOH (10 mL) and water (4 mL) afforded (S_a,S)-(+)-**3f** (0.9969 g, 78% over three steps) after chromatography (petroleum ether/EtOAc = 20:1–5:1). Liquid; $[\alpha]_D^{20} = +62.8$ ($c = 1.00$ in $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 11.48$ (brs, 1H; COOH), 5.86–5.70 (m, 1H; CH=), 5.23–5.02 (m, 4H; =CH_2 and 2×CH=), 2.71–2.57 (m, 1H; CH), 2.50–2.30 (m, 2H; CH_2), 2.28–2.08 (m, 2H; CH_2), 2.04–1.90 (m, 2H; CH_2), 1.44–1.24 (m, 4H; 2×CH_2), 0.90 ppm (t, $J = 6.9$ Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 202.9$, 179.5, 135.7, 117.1, 93.5, 93.4, 39.1, 38.5, 34.5, 31.2, 28.6, 22.2, 13.9 ppm; IR (neat): $\tilde{\nu} = 2959$, 2928, 1963, 1709, 1641, 1438, 1297, 1238, 1181 cm^{-1}; MS: m/z (%): 208 $[M]^+$ (0.27), 207 $[M-H]^+$ (0.82), 105 (100); elemental analysis calcd (%) for $C_{13}H_{20}O_2$: C 74.96, H 9.68; found: C 74.57, H 9.88.

Compound (S_a,S)-(+)-***3g***: Following method a, the reaction of (S_a,R)-2,6-dimethylhepta-3,4-dien-1-ol (0.1838 g, 1.31 mmol) and *p*-TsCl (0.7632 g, 4.00 mmol) in pyridine (3 mL) afforded the tosylate, which was used in the next step without further purification.

The reaction of the tosylate (prepared above) and NaCN (0.0972 g, 1.98 mmol) in DMSO (3 mL) afforded 3-allyldeca-4,5-dienenitrile after chromatography (petroleum ether/Et_2O = 100:1), which was used directly in the next step.

The reaction of 3-methylocta-4,5-dienenitrile (prepared above) with NaOH (0.5133 g, 12.8 mmol) in EtOH (2.5 mL) and water (0.8 mL), afforded (S_a,S)-(+)-**3g** (0.1552 g, 70% over three steps) after chromatography (petroleum ether/EtOAc = 5:1). Liquid; $[\alpha]_D^{20} = +83.5$ ($c = 1.10$ in $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 11.04$ (brs, 1H; COOH), 5.26–5.17 (m, 2H; 2×CH=), 2.75–2.58 (m, 1H; CH), 2.45 (dd, $J = 15.5$, 7.1 Hz, 1H; one proton of CH_2), 2.36–2.18 (m, 2H), 1.09 (d, $J = 6.6$ Hz, 3H; CH_3), 1.00 ppm (d, $J = 6.9$ Hz, 6H; 2×CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 200.9$, 179.4, 100.8, 96.8, 41.1, 29.8, 27.9, 22.4, 22.3, 20.3 ppm; IR (neat): $\tilde{\nu} = 2962$, 2929, 2871, 1960, 1711, 1458, 1410, 1293, 1229, 1199 cm^{-1}; MS: m/z (%): 168 $[M]^+$ (10.44), 111 (100); HRMS: m/z calcd for $C_{10}H_{16}O_2$: 168.1150 $[M]^+$; found: 168.1153.

*Compound (R)-**3h***: Phenylmagnesium bromide (50 mL, 1.0 M in THF, 50 mmol) was added dropwise within 30 min to a mixture of LiBr (1.4022 g, 16.1 mmol), CuBr·Me_2S (3.2787 g, 15.9 mmol), Me_2S (8 mL), and (3*S*,4*S*)-**11** (1.1165 g, 10.2 mmol) in THF (60 mL) at −50 °C under N_2. The reaction was stirred at −50 °C for an additional 3.5 h, then quenched with a saturated aqueous solution of NH_4Cl solution (20 mL) and acidified with HCl (5% aq) to pH ≈ 1. The resultant mixture was extracted with Et_2O (3×100 mL) and the combined organic extracts were washed with brine, dried over anhydrous Na_2SO_4, filtered, and concentrated under vacuum. The residue was purified by flash chromatography on silica gel (petroleum ether/EtOAc = 15:1–2:1) to afford β-allenoic acid, which was used directly in the next step.

A solution of β-allenoic acid (prepared above) in Et_2O (10 mL) was added dropwise within 12 min to a mixture of $LiAlH_4$ (0.5230 g, 13.8 mmol) in Et_2O (20 mL) at 0 °C under N_2. The reaction mixture was allowed to warm to RT and stirred at RT for an additional 1.8 h. After complete conversion of the starting material, monitored by TLC, the reaction mixture was quenched with water (3 mL) at 0 °C, dissolved in HCl (5% aq), extracted with Et_2O (3×100 mL), washed with a saturated aqueous solution of $NaHCO_3$ then brine, dried over anhydrous Na_2SO_4, filtered, and concentrated under vacuum. The residue was purified by flash chromatography on silica gel (petroleum ether/EtOAc = 10:1) to afford (R_a,S)-**12** [1.1236 g, 64% over two steps, d.r. = 11:1 (determined by inverse-gated-decoupling ^{13}C NMR analysis)]. Liquid; $[\alpha]_D^{20} = -291.7$ ($c = 0.70$ in $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 7.33–7.25$ (m, 4H; Ar−H), 7.23–7.14 (m, 1H; Ar−H), 6.21 (dd, $J = 6.5$, 2.4 Hz, 1H; CH=), 5.59 (t, $J = 6.5$ Hz, 1H; CH=), 3.57 (d, $J = 6.3$ Hz, 2H; CH_2), 2.57–2.43 (m, 1H; CH), 1.95 (brs, 1H; OH), 1.09 ppm (d, $J = 6.6$ Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 204.5$, 134.4, 128.6, 126.8, 126.5, 97.6, 95.8, 67.4, 36.4, 16.5 ppm; IR (neat): $\tilde{\nu} = 3349$, 3029, 2961, 2928, 2872, 1948, 1598, 1493, 1457, 1030 cm^{-1}; MS: m/z (%): 174 $[M]^+$ (52.08), 128 (100); HRMS: m/z calcd for $C_{12}H_{14}O$: 174.1045 $[M]^+$; found: 174.1049.

Following method a, the reaction of (R_a,S)-**12** (1.0636 g, 6.1 mmol) and *p*-TsCl (3.5204 g, 18.4 mmol) in pyridine (6 mL), afforded the tosylate, which was used in the next step without further purification.

The reaction of the tosylate (prepared above) and NaCN (0.4602, 9.4 mmol) in DMSO (8 mL) afforded (R_a,R)-**13** [0.6734 g, 60% over two steps, d.r. = 11:1 (determined by inverse-gated-decoupling ^{13}C NMR analysis)] after chromatography (petroleum ether/EtOAc = 30:1–20:1). Liquid; $[\alpha]_D^{20} = -260.5$ ($c = 0.98$ in $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 7.35–7.25$ (m, 4H; Ar−H), 7.25–7.16 (m, 1H; Ar−H), 6.31 (dd, $J = 6.3$, 3.0 Hz, 1H; CH=), 5.61 (t, $J = 6.2$ Hz, 1H; CH=), 2.75–2.60 (m, 1H; CH), 2.48–2.30 (m 2H; CH_2), 1.24 ppm (d, $J = 6.6$ Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 203.7$, 133.6, 128.5, 127.1, 126.6, 118.4, 98.1, 97.4, 30.5, 24.1, 19.8 ppm; IR (neat): $\tilde{\nu} = 3031$, 2967, 2930, 2247, 1950, 1598, 1495, 1457, 1421, 1380, 1337, 1263, 1071, 1027 cm^{-1}; MS: m/z (%): 182 $[M]^+$ (38.69), 115 (100); HRMS: m/z calcd for $C_{13}H_{13}N$: 183.1048 $[M]^+$; found: 183.1050.

The reaction of (R_a,R)-**13** (0.6573 g, 3.6 mmol) with NaOH (1.3207 g, 33.0 mmol) in EtOH (6 mL) and water (2.5 mL) afforded (*R*)-**3h** [0.5191 g, 72%, d.r. = 1:1 (determined by inverse-gated-decoupling ^{13}C NMR analysis)] after chromatography (petroleum ether/EtOAc = 20:1–10:1–2:1). Liquid; $[\alpha]_D^{20} = -26.1$ ($c = 0.88$ in $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 11.08$ (brs, 1H; COOH), 7.33–7.23 (m, 4H; Ar−H), 7.23–7.13 (m, 1H; Ar−H), 6.26–6.19 (m, 1H; CH=), 5.68–5.60 (m, 1H; CH=), 2.90–2.74 (m, 1H; CH), 2.58–2.30 (m 2H; CH_2), 1.22–1.13 ppm (m, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 203.8$, 178.9, 178.8, 134.4, 131.5, 128.6, 128.5, 126.90, 126.89, 126.6, 99.7, 99.6, 96.9, 96.7, 41.0, 40.9, 30.1, 29.9, 20.4, 20.2 ppm; IR (neat): $\tilde{\nu} = 3061$, 3031, 2965, 2928, 1949, 1709, 1598, 1493, 1456, 1413, 1291, 1231, 1198, 1071, 1028 cm^{-1}; MS: m/z (%): 202 $[M]^+$ (38.69), 160 (100); HRMS: m/z calcd for $C_{13}H_{14}O_2$: 202.0994 $[M]^+$; found: 202.0993.

*Compound (S)-**3h***: Following method a, the reaction of (S_a,R)-**12** (1.3789 g, 7.9 mmol) and *p*-TsCl (4.6465 g, 24.3 mmol, added in two portions) in pyridine (8 mL) afforded the tosylate, which was used in the next step without further purification.

The reaction of the tosylate (prepared above) and NaCN (1.1773 g, 24.0 mmol) in DMSO (8 mL) afforded 3-allyldeca-4,5-dienenitrile after chromatography (petroleum ether/EtOAc = 20:1), which was used directly in the next step.

The reaction of 3-allylocta-4,5-dienenitrile (prepared above) and NaOH (1.3011 g, 33.0 mmol) in EtOH (6 mL) and water (2.5 mL) afforded (*S*)-**3h** [0.4373 g, 27%, purity = 91% (determined by ^{1}H NMR analysis), d.r. = 1:1 (determined by inverse-gated-decoupling ^{13}C NMR analysis)] after chromatography (petroleum ether/EtOAc = 10:1–3:1). Liquid;

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$[\alpha]_D^{20}=+19.8$ ($c=0.97$ in $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): $\delta=11.73$ (br s, 1H; COOH), 7.34–7.13 (m, 5H; Ar–H), 6.26–6.18 (m, 1H; CH=), 5.68–5.58 (m, 1H; CH=), 2.90–2.74 (m, 1H; CH), 2.58–2.26 (m 2H; CH_2), 1.20–1.08 ppm (m, 3H; CH_3); IR (neat): $\tilde{\nu}=3031$, 2966, 2929, 1949, 1709, 1598, 1493, 1456, 1413, 1290, 1231, 1200, 1071, 1028 cm^{-1}; MS: *m/z* (%): 202 $[M]^+$ (14.62), 160 (100); HRMS: *m/z* calcd for $C_{13}H_{14}O_2$: 202.0994 $[M]^+$; found: 202.0995.

Synthesis of optically active butyrolactones (5) by the iodolactonization of 4-allenoic acids

*Method b, [(S)-(Z)-**5a**]*: Cs_2CO_3 (67.4 mg, 0.21 mmol), (R_a)-**3a** (38.9 mg, 0.20 mmol), and CH_2Cl_2 (5 mL) were added to a Schlenk tube at RT. The reaction mixture was stirred for 1.5 h at RT. Compound **8** (128.0 mg, 0.40 mmol) was added to the reaction mixture at −60 °C. After complete conversion of the starting material, monitored by TLC, the reaction mixture was quenched with water (3 mL) and a saturated aqueous solution of $Na_2S_2O_3$ (2 mL), then extracted with Et_2O (3×15 mL), washed with brine, dried over anhydrous Na_2SO_4, filtered, evaporated under vacuum. Purification by column chromatography on silica gel (petroleum ether/EtOAc=8:1) afforded (*S*)-(*Z*)-**5a** [54.1 mg, 85%, 98% *ee*, (*S*)-(*Z*)-**5a**/(*R*)-(*E*)-**5a**=97:3 (determined by ^1H NMR analysis of the crude reaction mixture before separation)]. Liquid; HPLC: OJ-H column, rate=0.8 mL min^{-1}, hexane/*i*PrOH=90:10, $\lambda=254$ nm, retention time (t_R)=13.5 [(*R*)-(*Z*)-**5a**, minor], 14.6 [(*S*)-(*Z*)-**5a**, major], 16.8 [(*R*)-(*E*)-**5a**, major], 27.6 min [(*S*)-(*E*)-**5a**, minor]; $[\alpha]_D^{20}=+17.0$ ($c=0.62$ in $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): $\delta=6.45$ (t, $J=7.8$ Hz, 0.03H; CH=, (*R*)-(*E*)-**5a**), 6.05 (td, $J=6.9$, 0.6 Hz, 0.96H; CH=, (*S*)-(*Z*)-**5a**), 4.90 (t, $J=7.5$ Hz, 0.04H; CH, (*R*)-(*E*)-**5a**), 4.80 (t, $J=6.9$ Hz, 0.96H; CH, (*S*)-(*Z*)-**5a**), 2.74–2.36 (m, 3H), 2.26–2.11 (m, 3H), 1.49–1.18 (m, 8H; 4×CH_2), 0.89 ppm (t, $J=6.8$ Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$, (*S*)-(*Z*)-**5a**): $\delta=176.5$, 138.6, 106.9, 84.3, 35.5, 31.6, 28.8, 28.6, 28.1, 27.9, 22.5, 14.1 ppm; IR (neat): $\tilde{\nu}=2955$, 2926, 2856, 1784, 1457, 1317, 1181, 1022 cm^{-1}; MS: *m/z* (%): 322 $[M]^+$ (31.94), 111 (100); HRMS: *m/z* calcd for $C_{12}H_{19}O_2I$: 322.0430 $[M]^+$; found: 322.0430.

*Compound (R)-(Z)-**5a***: Following method b, the reaction of Cs_2CO_3 (328.1 mg, 1.01 mmol), (S_a)-**3a** (196.0 mg, 1.00 mmol), and **8** (587.6 mg, 1.82 mmol, added in two portions) in CH_2Cl_2 (20 mL) afforded (*R*)-(*Z*)-**5a** [240.7 mg, 75%, 97% *ee*, (*R*)-(*Z*)-**5a**/(*S*)-(*E*)-**5a**=96:4 (determined by ^1H NMR analysis of the crude reaction mixture before separation)] after chromatography (petroleum ether/EtOAc=8:1). Liquid, HPLC: OJ-H column, rate=0.8 mL min^{-1}, hexane/*i*PrOH=90:10, $\lambda=254$ nm, $t_R=13.3$ [(*R*)-(*Z*)-**5a**, major], 14.9 [(*S*)-(*Z*)-**5a**, minor], 17.0 [(*R*)-(*E*)-**5a**, minor], 28.7 min [(*S*)-(*E*)-**5a**, major); $[\alpha]_D^{20}=-16.8$ ($c=1.08$ in $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): $\delta=6.45$ (t, $J=7.8$ Hz, 0.04H; CH=, (*S*)-(*E*)-**5a**), 6.06 (t, $J=6.8$, 0.97H; CH=, (*R*)-(*Z*)-**5a**), 4.90 (t, $J=7.4$ Hz, 0.04H; CH, (*S*)-(*E*)-**5a**), 4.80 (t, $J=6.9$ Hz, 0.96H; CH, (*R*)-(*Z*)-**5a**), 2.75–2.35 (m, 3H), 2.26–2.10 (m, 3H), 1.49–1.20 (m, 8H; 4×CH_2), 0.89 ppm (t, $J=6.6$ Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$, (*R*)-(*Z*)-**5a**): $\delta=176.4$, 138.5, 106.9, 84.2, 35.5, 31.5, 28.7, 28.6, 28.0, 27.8, 22.5, 14.0 ppm; IR (neat): $\tilde{\nu}=2954$, 2926, 2856, 1785, 1639, 1457, 1419, 1316, 1180, 1022 cm^{-1}; MS: *m/z* (%): 322 $[M]^+$ (35.02), 111 (100); HRMS: *m/z* calcd for $C_{12}H_{19}O_2I$: 322.0430 $[M]^+$; found: 322.0423.

*Compound (S)-(Z)-**5b***: Following method b, the reaction of Cs_2CO_3 (67.2 mg, 0.21 mmol), (R_a)-**3b** (42.0 mg, 0.20 mmol), and **8** (130.6 mg, 0.40 mmol) in CH_2Cl_2 (5 mL) afforded (*S*)-(*Z*)-**5b** [49.9 mg, 74%, 99% *ee*, (*S*)-(*Z*)-**5b**/(*R*)-(*E*)-**5b**=96:4 (determined by ^1H NMR analysis of the crude reaction mixture before separation)] after chromatography (petroleum ether/EtOAc/CH_2Cl_2=10:1:1). Liquid; HPLC: OJ-H column, rate=0.5 mL min^{-1}, hexane/*i*PrOH=90:10, $\lambda=230$ nm, $t_R=16.6$ [(*R*)-(*Z*)-**5b**, minor], 17.8 [(*S*)-(*Z*)-**5b**, major], 21.0 [(*R*)-(*E*)-**5b**, major], 31.3 min [(*S*)-(*E*)-**5b**, minor]; $[\alpha]_D^{20}=+10.8$ ($c=0.98$ in $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): $\delta=6.45$ (t, $J=8.0$ Hz, 0.04H; CH=, (*R*)-(*E*)-**5b**), 6.05 (t, $J=6.8$ Hz, 0.96H; CH=, (*S*)-(*Z*)-**5b**), 4.90 (t, $J=7.5$, 0.05H; CH, (*R*)-(*E*)-**5b**), 4.80 (t, $J=7.1$ Hz, 0.95H; CH, (*S*)-(*Z*)-**5b**), 2.75–2.35 (m, 3H), 2.26–2.10 (m, 3H), 1.51–1.18 (m, 10H; 5×CH_2), 0.89 ppm (t, $J=6.5$ Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$, (*S*)-(*Z*)-**5b**): $\delta=176.4$, 138.5, 106.9, 84.3, 35.5, 31.7, 29.1, 29.0, 28.6, 28.0, 27.9, 22.6, 14.1 ppm; IR (neat): $\tilde{\nu}=2954$, 2925, 2855, 1785, 1637, 1458, 1419, 1317, 1181, 1024 cm^{-1}; MS: *m/z* (%): 336 $[M]^+$ (5.63), 55 (100); HRMS: *m/z* calcd for $C_{13}H_{21}O_2I$: 336.0586 $[M]^+$; found: 336.0585.

*Compound (S)-(Z)-**5c***: Following method b, the reaction of Cs_2CO_3 (65.2 mg, 0.20 mmol), (R_a)-**3c** (50.0 mg, 0.20 mmol), and **8** (124.3 mg, 0.38 mmol) in CH_2Cl_2 (5 mL) afforded (*S*)-(*Z*)-**5c** [46.8 mg, 74%, 92% *ee*, (*S*)-(*Z*)-**5c**/(*R*)-(*E*)-**5c**=96:4 (determined by ^1H NMR analysis of the crude reaction mixture before separation)] after chromatography (petroleum ether/EtOAc=10:1). Liquid, HPLC: OJ-H column, rate=0.8 mL min^{-1}, hexane/*i*PrOH=90:10, $\lambda=254$ nm, $t_R=10.2$ [(*R*)-(*Z*)-**5c**, minor], 10.8 [(*S*)-(*Z*)-**5c**, major], 13.1 [(*R*)-(*E*)-**5c**, major], 16.2 min [(*S*)-(*E*)-**5c**, minor]; $[\alpha]_D^{20}=+14.8$ ($c=1.09$ in $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): $\delta=6.43$ (t, $J=7.8$ Hz, 0.04H; CH=, (*R*)-(*E*)-**5c**), 6.04 (td, $J=6.8$, 0.75 Hz, 0.98H; CH=, (*S*)-(*Z*)-**5c**), 5.88–5.72 (m, 1H; CH=), 5.04–4.85 (m, 2H; CH_2=), 4.78 (t, $J=6.9$ Hz, 1H; CH), 2.74–2.34 (m, 3H), 2.24–2.09 (m, 3H), 2.08–1.98 (m, 2H; CH_2), 1.48–1.20 ppm (m, 12H; 6×CH_2); ^{13}C NMR (75 MHz, $CDCl_3$, (*S*)-(*Z*)-**5c**): $\delta=176.4$, 139.1, 138.5, 114.1, 106.9, 84.3, 35.5, 33.7, 29.3, 29.1, 29.0, 28.8, 28.6, 28.0, 27.9 ppm; IR (neat): $\tilde{\nu}=3074$, 2925, 2853, 1786, 1639, 1457, 1419, 1317, 1181, 1023 cm^{-1}; MS (ESI): *m/z*: 377 $[M+H]^+$, 394 $[M+H_2O]^+$; HRMS: *m/z* calcd for $C_{16}H_{25}O_2INa$: 399.0791 $[M+Na]^+$; found: 399.0795.

*Compound (4R,5S)-(Z)-**5d***: Following method b, the reaction of Cs_2CO_3 (326.0 mg, 1.00 mmol), (R_a,*R*)-**3d** (153.1 mg, 0.99 mmol), and **8** (642.5 mg, 1.99 mmol, added in three portions) in CH_2Cl_2 (25 mL) afforded (4*R*,5*S*)-(*Z*)-**5d** [194.6 mg, 70%, 99% *ee*, (4*R*,5*S*)-(*Z*)-**5d**/(4*R*,5*R*)-(*E*)-**5d**=97:3 (determined by ^1H NMR analysis of the crude reaction mixture before separation)] after chromatography (petroleum ether/EtOAc=10:1). Liquid, HPLC: AD-H column, rate=0.8 mL min^{-1}, hexane/*i*PrOH=95:5, $\lambda=230$ nm, $t_R=8.8$ [(4*S*,5*R*)-(*Z*)-**5d**, minor], 9.4 min [(4*R*,5*S*)-(*Z*)-**5d**, major]; $[\alpha]_D^{20}=-32.5$ ($c=0.97$ in $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): $\delta=6.08$ (td, $J=6.8$, 1.2 Hz, 1H; CH=), 5.04 (d, $J=5.7$ Hz, 1H; CH), 2.98–2.83 (m, 1H; CH), 2.77 (dd, $J=17.0$, 8.0 Hz, 1H; one proton of CH_2), 2.41 (dd, $J=17.1$, 3.6 Hz, 1H; one proton of CH_2), 2.22 (quintet, $J=7.4$ Hz, 2H; CH_2), 1.03 (t, $J=7.5$ Hz, 3H; CH_3), 0.99 ppm (d, $J=6.9$ Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta=176.0$, 138.0, 98.8, 86.4, 37.2, 33.0, 29.1, 13.8, 12.6 ppm; IR (neat): $\tilde{\nu}=2968$, 2932, 2875, 1785, 1648, 1458, 1420, 1384, 1297, 1251, 1210, 1157, 1081, 1011 cm^{-1}; MS: *m/z* (%): 280 $[M]^+$ (44.51), 83 (100); HRMS: *m/z* calcd for $C_9H_{13}O_2I$: 279.9960 $[M]^+$; found: 279.9970.

*Compound (4R,5S)-(Z)-**5e***: Following method b, the reaction of Cs_2CO_3 (329.3 mg, 1.01 mmol), (R_a,*R*)-**3e** (181.6 mg, 1.00 mmol), and **8** (615.0 mg, 1.90 mmol, added in three portions) in CH_2Cl_2 (25 mL) afforded (4*R*,5*S*)-(*Z*)-**5e** [218.4 mg, 71%, 99% *ee*, (4*R*,5*S*)-(*Z*)-**5e**/(4*R*,5*R*)-(*E*)-**5e**=98:2 (determined by ^1H NMR analysis of the crude reaction mixture before separation)] after chromatography (petroleum ether/EtOAc=20:1). Liquid; HPLC: AD-H column, rate=0.8 mL min^{-1}, hexane/*i*PrOH=95:5, $\lambda=230$ nm, $t_R=7.8$ [(4*S*,5*R*)-(*Z*)-**5e**, minor], 8.7 min [(4*R*,5*S*)-(*Z*)-**5e**, major]; $[\alpha]_D^{20}=-25.9$ ($c=0.95$ in $CHCl_3$); ^1H NMR (300 MHz, $CDCl_3$): $\delta=6.07$ (td, $J=6.9$, 1.5 Hz, 1H; CH=), 5.04 (dd, $J=6.0$, 1.1 Hz, 1H; CH), 2.98–2.83 (m, 1H; CH), 2.77 (dd, $J=17.0$, 8.0 Hz, 1H; one proton of CH_2), 2.41 (dd, $J=16.8$, 3.6 Hz, 1H; one proton of CH_2), 2.27–2.15 (m, 2H; CH_2), 1.48–1.24 (m, 4H; 2×CH_2), 0.96 (d, $J=7.2$ Hz, 3H; CH_3), 0.92 ppm (t, $J=7.2$ Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta=176.0$, 136.8, 99.3, 86.5, 37.2, 35.2, 33.1, 30.2, 22.2, 13.9, 13.8 ppm; IR (neat): $\tilde{\nu}=2959$, 2929, 2872, 1789, 1637, 1457, 1420, 1381, 1319, 1283, 1268, 1208, 1164 cm^{-1}; MS: *m/z* (%): 308 $[M]^+$ (100); HRMS: *m/z* calcd for $C_{11}H_{17}O_2I$: 308.0273 $[M]^+$; found: 308.0271.

*Compound (4R,5S)-(Z)-**5f***: Following method b, the reaction of Cs_2CO_3 (66.0 mg, 0.20 mmol), (R_a,*R*)-**3f** (41.6 mg, 0.20 mmol), and **8** (126.6 mg, 0.39 mmol) in CH_2Cl_2 (5 mL), afforded (4*R*,5*S*)-(*Z*)-**5f** [55.6 mg, 83%, >99% *ee*, (4*R*,5*S*)-(*Z*)-**5f**/(4*R*,5*R*)-(*E*)-**5f**>99:1 (determined by ^1H NMR analysis of the crude reaction mixture before separation)] after chromatography (petroleum ether/EtOAc=20:1). Liquid; HPLC: AD-H column, rate=0.8 mL min^{-1}, hexane/*i*PrOH=95:5, $\lambda=230$ nm, $t_R=7.7$ [(4*S*,5*R*)-(*Z*)-**5f**, minor], 8.6 min [(4*R*,5*S*)-(*Z*)-**5f**, major], the racemic compound (*Z*)-**5f** was obtained by mixing (4*R*,5*S*)-(*Z*)-**5f** and (4*S*,5*R*)-(*Z*)-**5f**; $[\alpha]_D^{20}=-26.4$ ($c=0.81$ in $CHCl_3$); ^1H NMR (500 MHz, $CDCl_3$): $\delta=6.09$ (td, $J=6.8$, 1.2 Hz, 1H; CH=), 5.74–5.63 (m, 1H; CH=), 5.13–5.04 (m, 3H; CH and CH_2=), 2.87–2.78 (m, 1H; CH), 2.65 (dd, $J=17.5$,

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8.0 Hz, 1H; one proton of CH_2), 2.59 (dd, J=17.5, 5.0 Hz, 1H; one proton of CH_2), 2.32–2.25 (m, 1H; one proton of CH_2), 2.22 (q, J= 7.2 Hz, 2H; CH_2), 1.95–1.86 (m, 1H; one proton of CH_2), 1.46–1.38 (m, 2H; CH_2), 1.38–1.31 (m, 2H; CH_2), 0.92 ppm (t, J=7.2 Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=175.8, 137.8, 134.8, 118.0, 99.4, 86.1, 37.8, 35.3, 34.2, 32.3, 30.2, 22.3, 13.9; IR (neat): $\tilde{\nu}$=3078, 2957, 2928, 2858, 1788, 1641, 1465, 1438, 1419, 1337, 1302, 1242, 1220, 1155 cm^{-1}; MS (ESI): m/z: 394 $[M+AcOH]^+$, 352 $[M+H_2O]^+$, 335 $[M+H]^+$, 207 $[M-I]^+$; HRMS: m/z calcd for $C_{13}H_{19}O_2INa$: 357.0322 $[M+Na]^+$; found: 357.0315.

*Compound (4S,5R)-(Z)-**5f***: Following method b, the reaction of Cs_2CO_3 (66.3 mg, 0.20 mmol), (S_a,S)-**3f** (42.2 mg, 0.20 mmol), and **8** (129.1 mg, 0.40 mmol) in CH_2Cl_2 (5 mL) afforded (4*S*,5*R*)-(*Z*)-**5f** [59.9 mg, 88%, >99% *ee*, (4*S*,5*R*)-(*Z*)-**5f**/(4*S*,5*R*)-(*E*)-**5f**>99:1 (determined by ^{1}H NMR analysis of the crude reaction mixture before separation)] after chromatography (petroleum ether/EtOAc=20:1). Liquid; HPLC: AD-H column, rate=0.8 mL min^{-1}, hexane/*i*PrOH=95:5, λ=230 nm, t_R=7.7 [(4*S*,5*R*)-(*Z*)-**5f**, major], 8.6 min [(4*R*,5*S*)-(*Z*)-**5f**, minor], the racemic compound (*Z*)-**5f** was obtained by mixing (4*R*,5*S*)-(*Z*)-**5f** and (4*S*,5*R*)-(*Z*)-**5f**; $[\alpha]_D^{20}$=+26.3 (c=1.09 in $CHCl_3$); ^{1}H NMR (500 MHz, $CDCl_3$): δ=6.09 (td, J=6.8, 1.0 Hz, 1H; CH=), 5.74–5.64 (m, 1H; CH=), 5.13–5.05 (m, 3H; CH and CH_2=), 2.87–2.78 (m, 1H; CH), 2.66 (dd, J=17.5, 8.0 Hz, 1H; one proton of CH_2), 2.59 (dd, J=17.5, 5.0 Hz, 1H; one proton of CH_2), 2.32–2.25 (m, 1H; one proton of CH_2), 2.22 (q, J= 7.0 Hz, 2H; CH_2), 1.95–1.86 (m, 1H; one proton of CH_2), 1.47–1.38 (m, 2H; CH_2), 1.38–1.30 (m, 2H; CH_2), 0.92 ppm (t, 3H; J=7.0 Hz, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=175.8, 137.8, 134.8, 117.9, 99.4, 86.0, 37.7, 35.3, 34.2, 32.3, 30.2, 22.2, 13.9 ppm; IR (neat): $\tilde{\nu}$=3077, 2957, 2928, 2858, 1787, 1641, 1465, 1419, 1337, 1302, 1241, 1155 cm^{-1}; MS (ESI): m/z: 352 $[M+NH_4]^+$; HRMS: m/z calcd for $C_{13}H_{19}INaO_2$: 357.0322 $[M+Na]^+$; found: 357.0324.

*Compound (4S,5R)-(Z)-**5g***: Following method b, the reaction of Cs_2CO_3 (66.1 mg, 0.20 mmol), (S_a,S)-**3g** (33.6 mg, 0.20 mmol), and **8** (128.5 mg, 0.40 mmol) in CH_2Cl_2 (5 mL) afforded (4*S*,5*R*)-(*Z*)-**5g** [39.7 mg, 68%, 99% *ee*, (4*S*,5*R*)-(*Z*)-**5g**/(4*S*,5*R*)-(*E*)-**5g**>99:1 (determined by ^{1}H NMR analysis of the crude reaction mixture before separation)] after chromatography (petroleum ether/EtOAc=10:1). Liquid; HPLC: AD-H column, rate=0.8 mL min^{-1}, hexane/*i*PrOH=97:3, λ=230 nm, t_R=8.8 [(4*S*,5*R*)-(*Z*)-**5g**, major], 9.5 min [(4*R*,5*S*)-(*Z*)-**5g**, minor]; $[\alpha]_D^{20}$=+27.2 (c=1.70 in $CHCl_3$); ^{1}H NMR (500 MHz, $CDCl_3$): δ=5.89 (d, J=8.7 Hz, 1H; CH=), 5.02 (d, J=5.4 Hz, 1H; CH), 2.97–2.83 (m, 1H; CH), 2.77 (dd, J=17.1, 7.8 Hz, 1H; one proton of CH_2), 2.70–2.53 (m, 1H; CH), 2.39 (dd, J=17.1, 3.6 Hz, 1H; one proton of CH_2), 1.03 (t, J=6.8 Hz, 6H; 2×CH_3), 0.97 ppm (d, J=7.2 Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=176.0, 142.9, 96.5, 86.3, 37.2, 35.4, 33.0, 21.5, 21.4, 13.8 ppm; IR (neat): $\tilde{\nu}$=2963, 2930, 2869, 1789, 1647, 1464, 1419, 1384, 1297, 1257, 1210, 1157, 1083 cm^{-1}; MS (EI): m/z (%): 294 $[M]^+$ (10.62), 125 (100); HRMS: m/z calcd for $C_{10}H_{15}IO_2$: 294.0117 $[M]^+$; found: 294.0113.

*Compound (4R,5R)-(Z)-**5h***: Following method b, the reaction of Cs_2CO_3 (66.0 mg, 0.20 mmol), (*R*)-**3h** (40.0 mg, 0.20 mmol), and **8** (127.3 mg, 0.39 mmol) in CH_2Cl_2 (5 mL) afforded (4*R*,5*R*)-(*Z*)-**5h** [35.1 mg, 54%, >99% *ee*, (4*R*,5*R*)-(*Z*)-**5h**/(4*R*,5*S*)-(*E*)-**5h** >99:1 (determined by ^{1}H NMR analysis of the crude reaction mixture before separation)] after chromatography (petroleum ether/EtOAc=10:1). Liquid; HPLC: AD-H column, rate=0.6 mL min^{-1}, hexane/*i*PrOH=95:5, λ=254 nm, t_R=19.0 [(4*S*,5*S*)-(*Z*)-**5h**, minor], 20.2 min [(4*R*,5*R*)-(*Z*)-**5h**, major], the racemic compound (*Z*)-**5h** was obtained by mixing (4*R*,5*R*)-(*Z*)-**5h** and (4*S*,5*S*)-(*Z*)-**5h**; $[\alpha]_D^{20}$=−9.5 (c=0.88 in $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): δ= 7.71 (s, 1H; CH=), 7.42–7.31 (m, 3H; Ar−H), 7.25–7.20 (m, 2H; Ar−H), 4.35 (d, J=7.8 Hz, 1H; CH), 2.85 (dd, J=17.3, 8.6 Hz, 1H; one proton of CH_2), 2.71–2.54 (m, 1H; CH), 2.30–2.17 (m, 1H; one proton of CH_2), 1.06 ppm (d, J=6.6 Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=175.4, 147.3, 136.6, 128.8, 128.4, 128.0, 105.3, 84.1, 38.0, 36.5, 16.6 ppm; IR (neat): $\tilde{\nu}$=3055, 3024, 2963, 2929, 2873, 1789, 1598, 1575, 1492, 1446, 1418, 1382, 1310, 1274, 1208, 1147, 1065 cm^{-1}; MS: m/z (%): 328 $[M]^+$ (100); HRMS: m/z calcd for $C_{13}H_{13}O_2I$: 327.9960 $[M]^+$; found: 327.9967.

*Compound (4S,5S)-(Z)-**5h***: Following method b, the reaction of Cs_2CO_3 (67.0 mg, 0.21 mmol), (*S*)-**3h** (40.6 mg, 0.20 mmol), and **8** (126.4 mg, 0.39 mmol) in CH_2Cl_2 (5 mL) afforded (4*S*,5*S*)-(*Z*)-**5h** [33.9 mg, 51%, 95% *ee*, (4*S*,5*S*)-(*Z*)-**5h**/(4*S*,5*R*)-(*E*)-**5h**>99:1 (determined by ^{1}H NMR analysis of the crude reaction mixture before separation)] after chromatography (petroleum ether/EtOAc=10:1). Liquid; HPLC: AD-H column, rate=0.6 mL min^{-1}, hexane/*i*PrOH=95:5, λ=254 nm, t_R=19.0 [(4*S*,5*S*)-(*Z*)-**5h**, major], 20.2 min [(4*R*,5*R*)-(*Z*)-**5h**, minor], the racemic compound (*Z*)-**5h** was obtained by mixing (4*R*,5*R*)-(*Z*)-**5h** and (4*S*,5*S*)-(*Z*)-**5h**; $[\alpha]_D^{20}$=+8.4 (c=0.72 in $CHCl_3$); ^{1}H NMR (300 MHz, $CDCl_3$): δ= 7.71 (s, 1H; CH=), 7.43–7.30 (m, 3H; Ar−H), 7.30–7.18 (m, 2H; Ar−H), 4.35 (d, J=7.5 Hz, 1H; CH), 2.85 (dd, J=17.1, 8.7 Hz, 1H; one proton of CH_2), 2.71–2.54 (m, 1H; CH), 2.23 (dd, J=17.3, 9.8 Hz, 1H; one proton of CH_2), 1.06 ppm (d, J=6.6 Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ=175.4, 147.3, 136.6, 128.7, 128.3, 128.0, 105.3, 84.1, 38.0, 36.5, 16.5 ppm; IR (neat): $\tilde{\nu}$=3055, 3024, 2963, 2929, 2874, 1789, 1598, 1574, 1492, 1446, 1418, 1382, 1310, 1274, 1252, 1208, 1148, 1065 cm^{-1}; MS: m/z (%): 328 $[M]^+$ (78.81), 131 (100); HRMS: m/z calcd for $C_{13}H_{13}O_2I$: 327.9960 $[M]^+$; found: 327.9968.

Preparation of natural γ-butyrolactones[4f]

*Method c, synthesis of **15***: TBAF (1.61 mL, 1M in THF, 1.61 mmol) was added to a solution of (4*R*,5*S*)-(+)-(*Z*)-**5d** (129.6 mg, 0.45 mmol, 99% *ee*) in DMF (6 mL). The resultant mixture was stirred at 25°C for 5 h. After the reaction was complete, monitored by TLC, it was quenched with H_2O (5 mL), acidified with HCl (5% aq) to pH 1, and extracted with Et_2O (3×15 mL). The combined organic layers were washed with brine, dried over anhydrous Na_2SO_4, filtered, and evaporated under vacuum. Chromatography on silica gel (petroleum ether/EtOAc=5:1) afforded (4*R*, 5*R*)-5-*n*-butynyl-4-methyl-4,5-dihydro-2(3*H*)-furanone, which was used immediately in the next step.

(4*R*, 5*R*)-5-*n*-Butynyl-4-methyl-4,5-dihydro-2(3*H*)-furanone (prepared above), EtOAc (12 mL), and 10% Pd/C (dry, 49.3 mg, 0.05 mmol, 0.1 equiv) were added sequentially to an autoclave. The resultant mixture was stirred under H_2 (25 atm) at RT for 23 h and filtered through a short column of silica gel with Et_2O to afford **15**[10] (64.4 mg, 89% over two steps, 99% *ee*). Liquid; GC: CP-Chirasil-DEX CB column (25 m× 0.25 mm×0.25 μm); carrier=N_2; injector=250°C; detector (FID, H_2)= 250°C; oven temperature=135°C; t_R=14.3 (minor), 14.5 min (major); $[\alpha]_D^{20}$=+72.6 (c=1.01 in $CHCl_3$) [lit[10b]: $[\alpha]_D^{25}$=+73.8 (c=0.15 in MeOH); lit[10a]: $[\alpha]_D^{25}$=+76; lit[10c]: $[\alpha]_D^{25}$=+74 (c=0.4 in $CHCl_3$)]; ^{1}H NMR (300 MHz, $CDCl_3$): δ=4.49–4.38 (m, 1H; CH), 2.70 (dd, J=16.8, 7.8 Hz, 1H; proton of CH_2), 2.64–2.50 (m, 1H; CH), 2.26–2.14 (m, 1H; one proton of CH_2), 1.74–1.20 (m, 6H; 3×CH_2), 1.02 (d, J=6.9 Hz, 3H; CH_3), 0.92 ppm (t, J=6.3 Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ= 177.0, 83.6, 37.5, 32.9, 29.5, 27.9, 22.5, 13.9, 13.8 ppm; IR (neat): $\tilde{\nu}$=2958, 2935, 2874, 1778, 1458, 1422, 1383, 1337, 1294, 1209, 1170, 1093, 1080 cm^{-1}; MS: m/z (%): 156 $[M]^+$ (1.77), 99 (100); HRMS: m/z calcd for $C_9H_{16}O_2$: 156.1150 $[M]^{<M+>}$; found: 156.1153.

*Compound **16***: Following method c, the reaction of (4*R*,5*S*)-(+)-(*Z*)-**5e** (191.4 mg, 0.62 mmol, 99% *ee*) and TBAF (1.9 mL, 1M in THF, 1.9 mmol) in DMF (10 mL) afforded (4*R*, 5*R*)-5-*n*-hexynyl-4-methyl-4,5-dihydro-2(3*H*)-furanone, which was used immediately in the next step.

The reaction of (4*R*, 5*R*)-5-*n*-hexynyl-4-methyl-4,5-dihydro-2(3*H*)-furanone (prepared above), EtOAc (15 mL), and 10% Pd/C (dry, 67.2 mg, 0.06 mmol, 0.1 equiv) under H_2 (25 atm) afforded **16**[9,10d,11] (106.5 mg, 93% over two steps, 98% *ee*). Liquid; GC: G-BP column (25 m× 0.25 mm×0.25 μm); carrier=N_2; injector=250°C; detector (FID, H_2)= 250°C; oven temperature=60–180°C (1°C min^{-1}); t_R=87.6 (major), 88.5 min (minor); $[\alpha]_D^{20}$=+56.7 (c=1.04 in $CHCl_3$) [lit[9]: $[\alpha]_D^{22}$=+59.8 (c=0.4 in $CHCl_3$); lit[11b]: $[\alpha]_D^{12}$=+56.7 (c=2.47 in $CHCl_3$)]; ^{1}H NMR (500 MHz, $CDCl_3$): δ=4.47–4.41 (m, 1H; CH), 2.70 (dd, J=17.3, 7.8 Hz, 1H; one proton of CH_2), 2.63–2.54 (m, 1H; CH), 2.20 (dd, J=17.0, 4.0 Hz, 1H; one proton of CH_2), 1.70–1.60 (m, 1H; one proton of CH_2), 1.56–1.46 (m, 2H; CH_2), 1.40–1.24 (m, 7H), 1.01 (d, J=7.0 Hz, 3H; CH_3), 0.89 ppm (t, J=6.8 Hz, 3H; CH_3); ^{13}C NMR (75 MHz, $CDCl_3$): δ= 177.0, 83.7, 37.5, 32.9, 31.6, 29.8, 29.0, 25.8, 22.5, 14.0, 13.8 ppm; IR (neat): $\tilde{\nu}$=2955, 2930, 2858, 1778, 1466, 1421, 1382, 1335, 1294, 1211, 1167, 1080 cm^{-1}; MS (ESI): m/z: 207 $[M+Na]^+$, 202 $[M+H_2O]^+$, 185 $[M+H]^+$; HRMS: m/z calcd for $C_{11}H_{20}O_2$: 184.1463 $[M]^+$; found: 184.1469.

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